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The EU's Strategic Dependence on Critical Raw Materials

EUROPEAN ECONOMY PROGRAMME

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Summary

- **Demand is rising for structural reasons.** Europe's energy transition and industrial policies increase mineral demand. International Energy Agency underlines that demand for minerals used in clean energy technologies will more than double by 2040.
- **Processing is often the weakest link.** Europe is often most exposed not at the mine extraction level, but in processing, conversion, and intermediate products needed for manufacturing.
- **China's leverage derives from scale and policy tools.** China's position is strongest in several processing and intermediate segments. China weaponises tools such as export controls to slow or reshape flows.
- **The EU has moved from elaboration to implementation.** The Critical Raw Materials Act entered into force on 23 May 2024, and the EU has started to implement it through "Strategic Projects", including 47 projects inside the EU and 13 in third countries. Further initiatives were developed by late 2025 in the context of RESourceEU.
- **Rules can improve resilience, but timing matters.** Compared with China's faster, more coordinated model, Europe risks turning regulatory credibility into a liability.

Why Critical Raw Materials matter

Since the Bronze Age, the use of minerals has been at the core of human progress[1]. Today in particular, critical raw materials can be described as the *lifeblood* of the modern digital economy and the green transition. More specifically, the market for critical minerals has grown rapidly over the past five years, reaching **$325 billion in 2023[2]**. This rising demand is driven largely by green technologies. For example, an electric vehicle requires roughly **six times more** critical minerals compared to a conventional one, with batteries alone requiring, on average, **200 kilograms** of critical minerals per vehicle[3]. Goldman Sachs forecasts that by **2030**, electric vehicles will account for **72%** of all new-vehicle sales in the EU[4] and **50%** in the United States, with the potential to reach **half of global sales by 2035**[5].

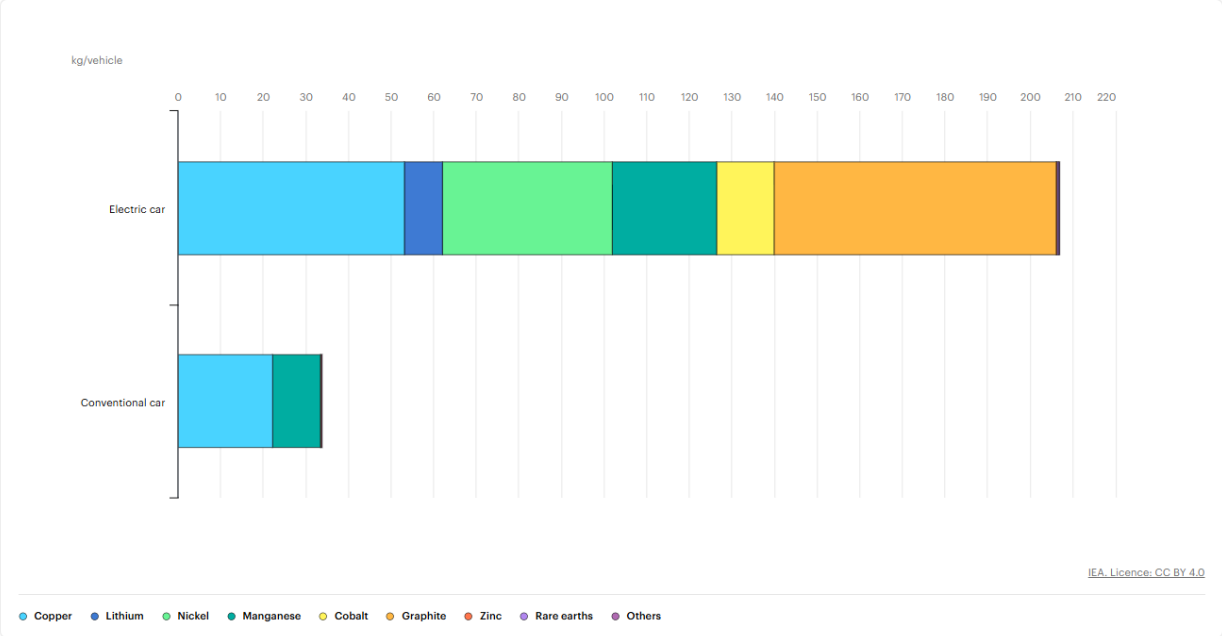


Figure 1: The elements required to produce an electric vs. a conventional vehicle [6]

At the same time, according to the U.S. Geological Survey (USGS), an average person needs[7] **431 kg of copper**, **1,221 kg of aluminium**, and **9,818 kg of iron ore** over the course of their lifetime[8]. As a result, the surging demand for minerals across many areas of life creates challenges for countries that lack domestic reserves.

[1] Amzallag, N. (2009). From metallurgy to Bronze Age civilizations: The synthetic theory. *American Journal of Archaeology, 113*(4), 497–519. https://doi.org/10.3764/aja.113.4.497
[2] International Energy Agency. (2024). *Outlook for key minerals*. In *Global Critical Minerals Outlook 2024*. https://www.iea.org/reports/global-critical-minerals-outlook-2024/outlook-for-key-minerals
[3] Similarly, a smartphone requires the use of about 30 different chemical elements.
[4] European Parliament. (2022). *EU ban on sale of new petrol and diesel cars from 2035 explained*. https://www.europarl.europa.eu/topics/en/article/20221019STO44572/eu-ban-on-sale-of-new-petrol-and-diesel-cars-from-2035-explained.
[5] Goldman Sachs. (2023). *Electric vehicles are forecast to be half of global car sales by 2035*. https://www.goldmansachs.com/insights/articles/electric-vehicles-are-forecast-to-be-half-of-global-car-sales-by-2035.
[6] International Energy Agency. (2021). *Minerals used in electric cars compared to conventional cars*. In *The role of critical minerals in clean energy transitions*. International Energy Agency. https://www.iea.org/data-and-statistics/charts/minerals-used-in-electric-cars-compared-to-conventional-cars
[7] Per-capita accounting estimate of the raw extraction needed to sustain a U.S.-level lifestyle across a lifetime (buildings, roads, bridges, vehicles, machinery, power and industrial infrastructure).
[8] U.S. Geological Survey. (2025). *How large is a lifetime supply of minerals for the average person? https://www.usgs.gov/faqs/how-large-a-lifetime-supply-minerals-average-person?*

In 2022, the USGS compiled a list of **50 critical minerals**[9], along with an additional subcategory of **17 rare earth elements (REEs)**[10], which are used in a wide array of products, including permanent magnets in many wind turbines and electric-vehicle motors, components of smartphones, and advanced defence platforms such as the F-35, which the U.S. Department of War states that contains more than 900 pounds of rare earth elements[11]. The aforementioned list was updated in 2025, adding 10 new critical minerals[12]. Likewise, the EU updated its list of critical raw materials in **2023** (following a revision in **2020**). This list includes **34 critical raw materials,** and for the first time, **17 strategic raw materials**, which constitute a subset of the critical raw materials[13]. These are considered essential for the strategic technologies used in the EU's green, digital, defence, and aerospace sectors. The EU uses the list to focus attention on the key raw materials needed by the European economy. The detailed table can be found in the appendix of this policy brief.

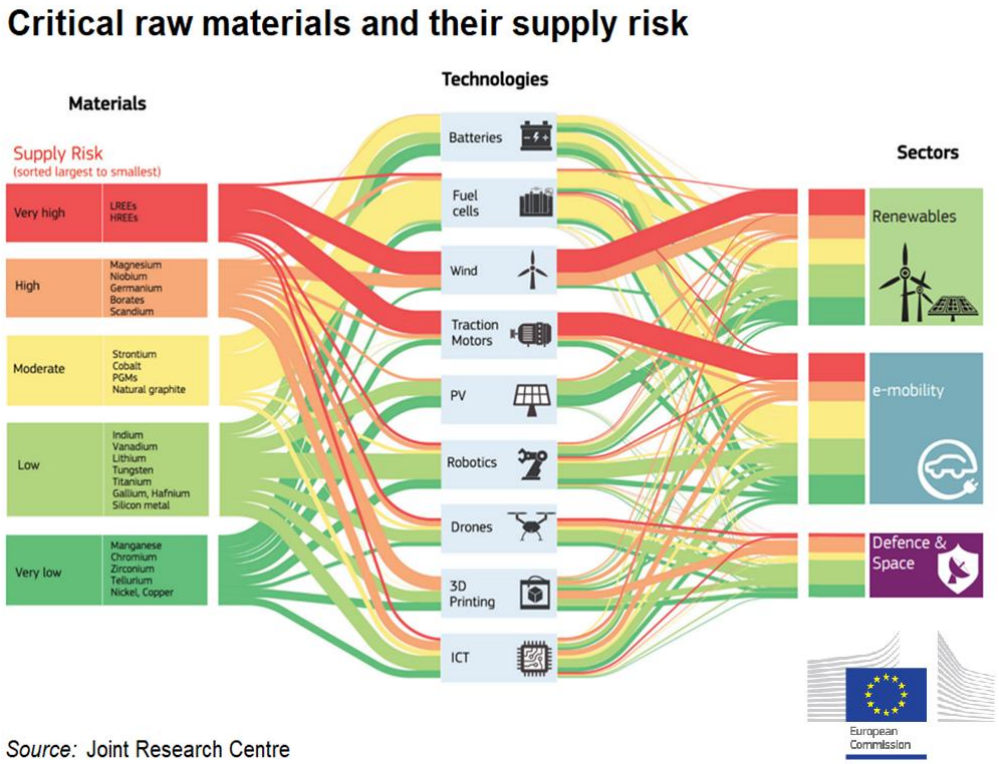


Figure 2: Critical raw materials and their supply risk[14]

Additionally, we observe that competition for global resources and particularly for raw materials is on the rise. Russia's war against Ukraine and the subsequent negotiation between the USA and Ukraine focusing on Ukraine's critical raw materials reserves[15], the American aspirations for Greenland's mineral riches[16] and the rising cross-

[9] United States Geological Survey. (2022). *U.S. Geological Survey releases 2022 list of critical minerals*. https://www.usgs.gov/news/national-news-release/us-geological-survey-releases-2022-list-critical-minerals.
[10] United States Geological Survey. *Rare earths statistics and information*. https://www.usgs.gov/centers/national-minerals-information-center/rare-earths-statistics-and-information.
[11] Lopez, C. T. (2024, March 11). *DOD looks to establish "mine-to-magnet" supply chain for rare earth materials*. *U.S. Department of War*. https://www.war.gov/News/News-Stories/Article/Article/3700059/dod-looks-to-establish-mine-to-magnet-supply-chain-for-rare-earth-materials/
[12] Mineral Resources Program. (2025). *About the 2025 list of critical minerals*. U.S. Geological Survey. https://www.usgs.gov/programs/mineral-resources-program/science/about-2025-list-critical-minerals
[13] European Commission. *Critical and strategic materials*. RMIS. https://rmis.jrc.ec.europa.eu/eu-critical-raw-materials.
[14] Eurostat. (2025). International trade in critical raw materials. Statistics Explained. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=International_trade_in_critical_raw_materials
[15] Debusmann Jr., B., & Bateman, T. (2025). *US and Ukraine sign long-awaited natural resources deal*. BBC News. https://www.bbc.com/news/articles/c5ypw7pn9q3o
[16] Murray, A. (2025). *The story behind the scramble for Greenland's rare earths*. BBC Worklife. https://www.bbc.com/worklife/article/20251104-the-story-behind-the-scramble-for-greenlands-rare-earths

strait tensions involving China and Taiwan, a critical hub in high-tech supply chains, especially semiconductors[17], underscore this trend. Many other mineral-rich developing countries are also pursuing resource-based industrial strategies, seeking to use domestic reserves to build industries in renewables, electric vehicles, and electronics. Indonesia, rich in nickel reserves, illustrates this shift, having imposed restrictions on nickel exports[18]. Even the EU-Mercosur trade deal included provisions about securing critical raw material supplies[19].

How the green transition threatens to build a new dependence for Europe

European Commission President Ursula von der Leyen warned about Europe's growing dependence on raw materials by stating in her 2022 State of the Union address that "Lithium and rare earths will soon become more important than oil and natural gas… We must avoid becoming dependent again, as we did with oil and gas"[20]. In a recent study, the **International Energy Agency (IEA)** expects the market for critical raw materials, necessary for green energy technologies, to **reach USD 770 billion by 2040**[21]. This finding underlines their strategic importance, especially when combined with the ongoing "politicisation" of cross-border supply chains amid the economic shock caused by the pandemic and rising global tensions. At the same time, the EU's drive to reduce energy dependence from third countries is tied to new dependencies. Scaling up wind, solar, and geothermal energy, as well as energy-storage systems, will require more than 3 billion tonnes of minerals and metals in order to keep global warming below 2°C[22]. A related Goldman Sachs analysis offers an interesting picture in the following figure[23].

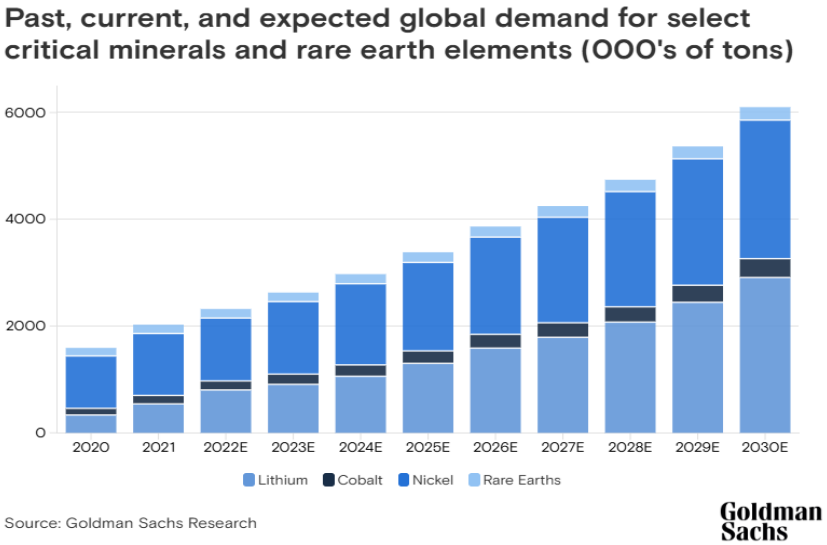


Figure 3: Past, current, and projected global demand for selected critical minerals and rare earth elements (in thousands of tonnes).[24]

[17] Brown, K. (2025). *How did semiconductors become so central to Taiwan's economic progress?* Economics Observatory. https://www.economicsobservatory.com/how-did-semiconductors-become-so-central-to-taiwans-economic-progress
[18] Hertanti, R. (2023). *Indonesia's export ban on raw minerals*. CETRI. https://www.cetri.be/Indonesia-s-export-ban-on-raw?lang=fr
[19] European Commission. (n.d.). *Factsheet: EU-Mercosur partnership agreement – Enhancing trade and investment in critical raw materials*. https://policy.trade.ec.europa.eu/eu-trade-relationships-country-and-region/countries-and-regions/mercosur/eu-mercosur-agreement/factsheet-eu-mercosur-partnership-agreement-enhancing-trade-and-investment-critical-raw-materials_en
[20] European Commission (2022). Critical Raw Materials Act: Securing the new gas & oil at the heart of our economy [Statement]. European Commission.
[21] Under IEA's scenario consistent with announced pledges/net-zero trajectory. Source: International Energy Agency. (2024). *Outlook for key minerals*. In *Global critical minerals outlook 2024*. International Energy Agency. https://www.iea.org/reports/global-critical-minerals-outlook-2024/outlook-for-key-minerals
[22] World Bank. (2019). *Climate-smart mining: Minerals for climate action*. https://www.worldbank.org/en/topic/extractiveindustries/brief/climate-smart-mining-minerals-for-climate-action
[23] Cohen, J., Shirley, W., & Svensson, K. (2023). *Resource realism: The geopolitics of critical mineral supply chains*. Goldman Sachs. Retrieved from https://www.goldmansachs.com/insights/articles/resource-realism-the-geopolitics-of-critical-mineral-supply-chains
[24] ibid

Therefore, the strategic reorientation of the green transition raises important questions about the new geo-economic balances and the implications this entails. For the European Commission, securing access to critical raw materials is a vital priority for maintaining the EU's competitiveness within the global economy[25]. This becomes entirely clear when we consider that the EU has limited reserves compared to its needs[26]. As a result, although mineral resources are spread across the world's continents, they are also concentrated in certain regions, which are often prone to conflict, instability, and broader risk, factors that can disrupt supply and hinder their movement through global markets[27].

The following map[28] shows the main suppliers of critical raw materials to the European Union, together with a colour scale indicating each country's level of governance (from high (green) to very low (red)). For each region, the map lists the specific materials that are exported and/or processed, as well as the corresponding share of EU imports. The map clearly illustrates how dependent the EU is on a range of external sources for key minerals and metals. It also highlights the link between the supply of these materials and the quality of governance in supplier countries and therefore the potential fragility of these trade relationships.

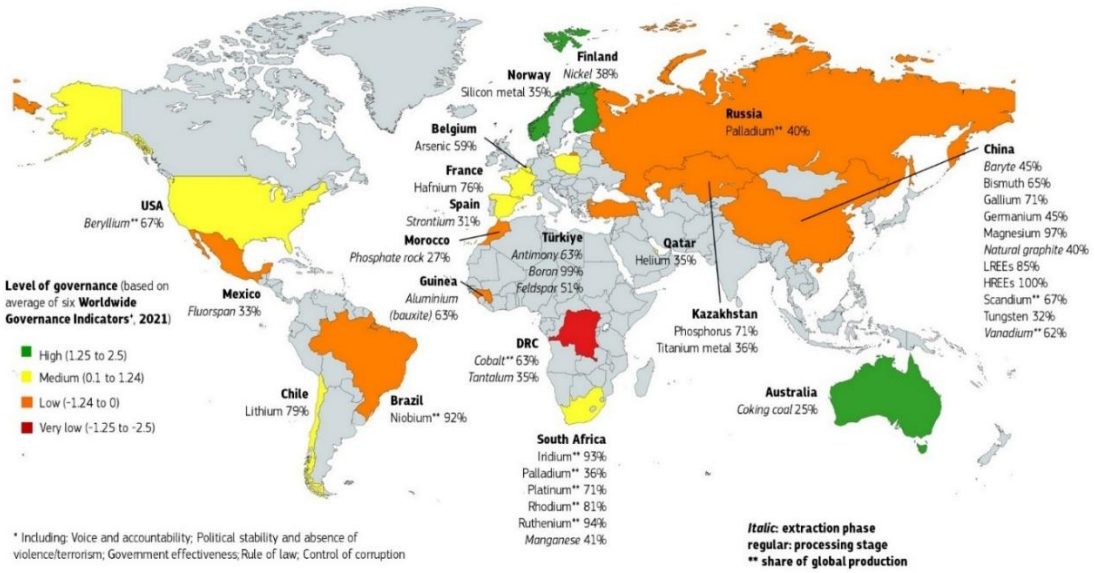


Figure 4: Main suppliers of critical raw materials to the EU (2023) and their level of governance [29].

In addition, another important point is that processing is carried out mainly, and for some resources almost exclusively, in China[30]. This creates a grave geopolitical risk, an asymmetric dependence that has generated multiple crises in the past and provides Beijing with an additional strategic instrument, the use of trade restrictions. As a result, Europe's dependence on critical raw materials should not be solely interpreted as an import problem. In the current strategic setting, dependence is better understood as **a problem of control**. Europe is vulnerable where a small number of actors can slow, raise the cost of, or condition Europe's access to key inputs, which Europe cannot replace them quickly. This is why the EU has begun to treat critical raw materials as part of its **economic security**, not only as a trade or industrial policy issue[31]. The central concern is that supply chains are increasingly

[25] European Commission. (2023). *Proposal for a regulation of the European Parliament and of the Council establishing a framework for ensuring a secure and sustainable supply of critical raw materials and amending Regulations (EU) 168/2013, (EU) 2018/858, 2018/1724 and (EU) 2019/1020* (COM(2023) 160 final). *EUR-Lex*. https://eur-lex.europa.eu/legal-content/EL/TXT/HTML/?uri=CELEX:52023PC0160

[26] European Round Table for Industry. (2023). *Expert paper: The EU Critical Raw Materials Act*. European Round Table for Industry. https://ert.eu/wp-content/uploads/2023/09/ERT-Expert-Paper-on-the-Critical-Raw-Materials-Act_September-2023.pdf

[27] International Energy Agency. (2023). *Clean energy supply chains vulnerabilities*. In *Energy Technology Perspectives 2023*. https://www.iea.org/reports/energy-technology-perspectives-2023/clean-energy-supply-chains-vulnerabilities.

[28] European Commission. Critical and strategic materials. RMIS. https://rmis.jrc.ec.europa.eu/eu-critical-raw-materials

[29] ibid

[30] ibid

[31] European Commission. (n.d.). Critical Raw Materials Act. https://single-market-economy.ec.europa.eu/sectors/raw-materials/areas-specific-interest/critical-raw-materials/critical-raw-materials-act_en

organised around **chokepoints**, controlled by strategic competitors. This means that Europe should not treat coercion risk as a rare scenario. In a more competitive EU–China relationship, it stands as a likely scenario.

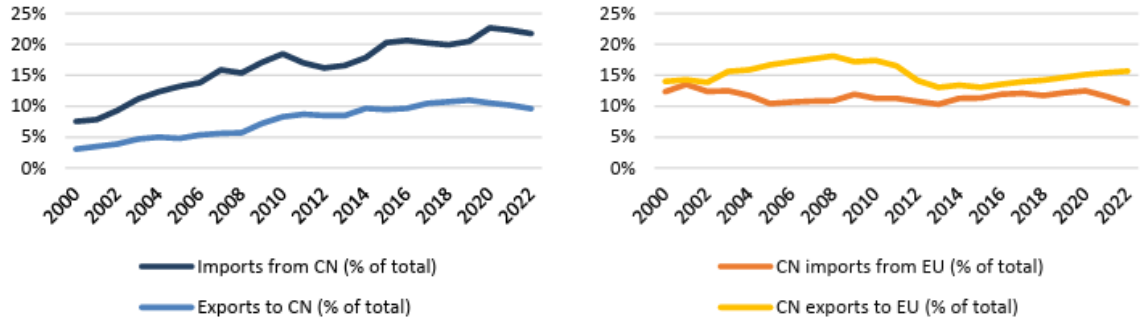


Figure 5: EU direct trade exposure to China (left; goods, 2000-2022), CN direct trade exposure to EU (right; goods, 2000-2022)[32]

China's Strategy for Critical Minerals Security

China has been developing its strategy on critical minerals for decades. As early as 1992, Chinese leader **Deng Xiaoping** recognised the strategic value of rare earths, remarking that "the Middle East has oil, China has rare earths" [33]. Through a combination of strategic investments, industrial policy, low labour costs, relatively lax environmental regulation and continuous engagement with states rich in resources, China consolidated global dominance in this sector[34].

Today, China is a leading player in the critical raw materials market, holding decisive leverage in **midstream/downstream processing and manufacturing** of **cobalt, copper, lithium, and nickel**, as well as in the production and refining of **rare earths**[35]. Western dependence on China is substantial, with the EU showing greater dependence than the United States[36]. China controls sizeable deposits of several key critical materials. It holds roughly **10%** of known global **lithium** reserves, about **28%** of **graphite**, nearly **49%** of **rare earth elements (REEs)**, and around **20%** of **zinc**. Beyond reserves, China's leverage becomes most pronounced in downstream processing. It represents about **44%** of worldwide **copper refining**, processes roughly **70–75%** of global **lithium** and **cobalt**, and accounts for **over 90%** of refining capacity for **REEs** and **battery-grade graphite**[37].

[32] Vandermeeren, F. (2024, January 17). Understanding EU-China exposure (Single Market Economics Papers). European Commission. https://doi.org/10.2873/425398 https://single-market-economy.ec.europa.eu/publications/understanding-eu-china-exposure_en

[33] Yiying Zhang, Guoyi Han, & Jürisoo, M. (2014). *The geopolitics of China's rare earths: a glimpse of things to come in a resource-scarce world?* Stockholm Environment Institute. http://www.jstor.org/stable/resrep00363

[34] Tse, Pui-Kwan, (2011), China's rare-earth industry: U.S. Geological Survey Open-File Report 2011–1042, 11 p., https://pubs.usgs.gov/of/2011/1042.

[35] Vivoda, Vlado & Matthews, Ron & McGregor, Neil. (2024). A Critical Minerals Perspective on the Emergence of Geopolitical Trade Blocs. Resources Policy. 89. 104587. 10.1016/j.resourpol.2023.104587.

[36] Müller, M. (2023). The 'new geopolitics' of mineral supply chains: A window of opportunity for African countries. *South African Journal of International Affairs*, *30*(2), 177–203. https://doi.org/10.1080/10220461.2023.2226108

[37] International Energy Agency. (2025). Global critical minerals outlook 2025. https://www.iea.org/reports/global-critical-minerals-outlook-2025

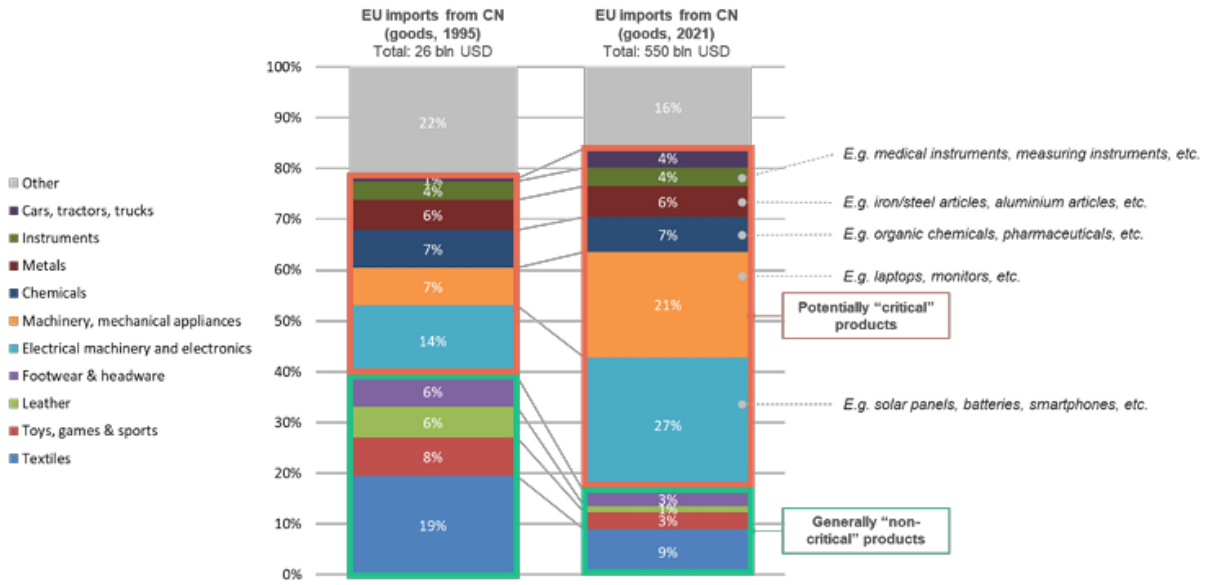


Figure 6: EU imports from China (goods, 1995-2021)[38]

Although new deposits of critical minerals continue to be discovered around the world, China remains the leading producer for most of them, partly because it extracts at a faster pace than other countries[39].

Chinese firms are actively involved across the entire supply chain, investing in mining projects worldwide[40]. They have developed strategic networks in selected countries to strengthen the organisational capacity of domestic companies through joint ventures and the transfer of international expertise[41]. Through an **"infrastructure-for-minerals" strategy**, China has secured access to resources by using state-owned banks and major business actors to provide loans to resource-rich countries, especially in **Africa** and **Latin America**, in exchange for mineral commodities[42]. From **2000 to 2017**, China provided at least **$18 billion** in loans for mining projects in Africa[43]. At the same time, the **Belt and Road Initiative (BRI)**, launched in **2013**, has reinforced infrastructure investment that facilitates the shipment of minerals to China[44], with improved connectivity through roads, ports, and railways supporting these flows[45].

As a result, if we examine China's tools for expanding its influence abroad, we can identify two main pathways. The first is the use of state-owned enterprises (SOEs). For example, China pursues an SOE-based strategy in Africa, leveraging the financial pressures and constraints facing its domestic mining sector. Through its SOEs, it offers financing and infrastructure investment in order to entrench itself in high-risk countries, while building close ties with local elites. For example, through the **China-Africa Development Fund**, Chinese state-owned firms have expanded into the Bushveld Complex in South Africa, a mineral-rich region that holds the world's largest reserves of **platinum-group metals** (important for catalytic converters) and the **third-largest** concentration of **vanadium** (critical for renewable energy, aerospace, and defence). Using state-backed resources, Chinese companies have

[38] Vandermeeren, F. (2024, January 17). Understanding EU-China exposure (Single Market Economics Papers). European Commission. https://doi.org/10.2873/425398
[39] U.S. Geological Survey. (2023). *Mineral commodity summaries 2023*. https://doi.org/10.3133/mcs2023
[40] Nakano, J. (2021). The geopolitics of critical minerals supply chains. Center for Strategic and International Studies. https://www.csis.org/analysis/geopolitics-critical-minerals-supply-chains
[41] Cooke, F. L., et al. (2015). Mining with a high-end strategy: A study of Chinese mining firms in Africa and human resources implications. International Journal of Human Resource Management, 26(21), 2843–2861. https://doi.org/10.1080/09585192.2015.1071863
[42] Mihalyi, D., Adam, A., & Hwang, J. (2020). *Resource-backed loans: Pitfalls and potential*. Natural Resource Governance Institute. https://resourcegovernance.org/publications/resource-backed-loans-pitfalls-and-potential
[43] Bräutigam, D., & Gallagher, K. P. (2014). Bartering globalization: China's commodity-backed finance in Africa and Latin America. *Global Policy, 5*(3), 346–356. https://doi.org/10.1111/1758-5899.12138
[44] Rahman, Z. U. (2022). A comprehensive overview of China's Belt and Road Initiative and its implications for the region and beyond. *Journal of Public Affairs, 22*(1), 3. https://doi.org/10.1002/pa.2298
[45] Ehizuelen, M. M. O. (2017). More African countries on the route: The positive and negative impacts of the Belt and Road Initiative. *Transnational Corporations Review, 9*(4), 356-373. https://doi.org/10.1080/19186444.2017.1401260

made eight major investments in the area, contributing to a shift in which minerals have become a central pillar of South Africa's export growth, with nearly 50% of these exports directed to China[46].

Secondly, **China makes use of private companies backed by state capital**. By acquiring stakes in local mining firms and financing smaller producers, it expands its presence while reducing concerns about direct foreign control over strategic resources, such as **niobium** in Brazil and **tantalum** in Australia. This approach is particularly visible in the three countries that account for around **90%** of global lithium production (**Chile, Argentina, and Australia**)[47].

When it comes to its national reserves, China has sharply increased state spending on geological exploration. Since 2022, annual spending has exceeded CNY 100 billion (around USD 14 billion), representing the highest funding in the last decade. Additionally, at the subnational level, at least half of China's 34 provincial-level administrative divisions, including major resource regions such as Xinjiang, have announced higher subsidies or access arrangements to accelerate mineral exploration[48]. Moreover, several legal reforms have reinforced this strategy. Under the 2024 revision of China's **Mineral Resources Law**, farmland can be requisitioned for the development of strategic mineral resources. In parallel, Beijing supports domestic mining through packages of **subsidies, tax benefits, and other policy support** [49].

Therefore, China's relevance to Europe's critical raw materials risk does not rest primarily on being a major miner of everything. It rests on China's position **across the value chain**, especially in processing and manufacturing ecosystems. In practical terms, China sits at conversion and refining nodes that turn raw inputs into industrially usable forms and intermediate products needed for clean-tech manufacturing. This position gives Beijing a form of **structural power** as it can shape the conditions under which other states access key inputs, even when global supply is not scarce. Their abundance in raw materials has enabled China to step up the use of export controls on critical minerals, treating them as an instrument of wider geopolitical leverage.

An early indication of this leverage emerged in 2010, when China's restrictions on rare-earth exports, in the context of its dispute with Japan[50], exposed this strategic vulnerability. In **2023** it restricted exports of **gallium** and **germanium**[51] and in **2025 as a response to US tariffs** China responded with export controls on key rare earths[52]. Additionally, in early **2026** China restricted exports of some rare earth elements and other items to **Japan**[53]. Among the aforementioned, we choose to further illustrate the case of 2025 as it is more related to the EU context.

In April 2025, China added seven rare earth elements to its export control list, requiring exporters to obtain licences[54]. These materials are vital for producing permanent magnets, essential for wind turbines and industrial applications. The European Commission opened bilateral contacts with Beijing immediately. In June 2025, it launched a portal through which manufacturers could report the status of their licence requests. If requests were urgent, the Commission forwarded the information to the Chinese authorities in an effort to accelerate approval.

[46] Lee, T. (2019). *Mining the future: China's grip on critical minerals and metals*. *Foreign Policy*. https://foreignpolicy.com/2019/05/01/mining-the-future-china-critical-minerals-metals/
[47] ibid
[48] White, E. (2025). China raises state funding for strategic minerals amid US trade war. Financial Times. https://www.ft.com/content/cace5b0f-e08c-4cb9-aac5-c3117d5a93bc
[49] International Energy Agency. (2025). Global critical minerals outlook 2025. https://www.iea.org/reports/global-critical-minerals-outlook-2025
[50] Bradsher, K. (2010). *China's monopoly on rare earths puts the world at risk*. *The New York Times*. https://www.nytimes.com/2010/09/23/business/global/23rare.html.
[51] Lv, A., & Goh, B. (2023). *U.S. firm AXT applying for permits after China restricts chipmaking exports*. Reuters. https://www.reuters.com/technology/us-firm-axt-applying-permits-after-china-restricts-chipmaking-exports-2023-07-04/.
[52] Jackson, L., Lv, A., Onstad, E., & Scheyder, E. (2025, April 4). China hits back at US tariffs with export controls on key rare earths. Reuters. https://www.reuters.com/world/china-hits-back-us-tariffs-with-rare-earth-export-controls-2025-04-04/
[53] Liu, J. (2026, January 6). Japanese PM's Taiwan comments prompt China to ban certain exports to Japan. CNN Business. https://edition.cnn.com/2026/01/06/business/china-japan-export-controls-intl-hnk
[54] European Parliament. (2025). *European Parliament resolution of 10 July 2025 on tackling China's critical raw materials export restrictions (2025/2800(RSP))* (P10_TA(2025)0166). https://www.europarl.europa.eu/doceo/document/TA-10-2025-0166_EN.html

However, the European Chamber of Commerce in China reported that, based on data from 22 European firms collected between August and early September 2025, only 19 of 141 licence applications had been approved[55].

The European Union's Strategy for Critical Raw Materials

Today, the EU is heavily dependent on imports of critical raw materials. While Europe has significant downstream industrial capacity, domestic extraction contributes marginally to global output of metals and minerals[56]. In **2023** the EU extracted **5.2 billion tonnes** (equivalent to **5%** of global material extraction), compared with a total worldwide production of **98.9 billion tonnes**[57].

Material extraction and consumption in EU and non-EU, 2023
(billion tonnes)

	serving EU's consumption		serving consumption in the non-EU rest of the world		Total extracted (DE*)	
extracted in EU	4.0	*3.8%*	1.2	*1.2%*	**5.2**	*5.0%*
extracted in the non-EU rest of the world	2.2	*2.1%*	96.8	*92.9%*	**98.9**	*95.0%*
Total consumed (RMC**)	**6.1**	*5.9%*	**98.0**	*94.1%*	**104.2**	*100%*

* domestic extraction; ** raw material consumption
Source: Eurostat (online data code: env_ac_rme), www.materialflows.net

eurostat

Figure 7: Material extraction and consumption in EU and non-EU, 2023 (billion tonnes) [58]

By contrast, China, as already noted, has entrenched a dominant position, especially in **refining**[59], leaving the EU with a serious dependence on Chinese processing capacity. In some cases, Europe's reliance on China for certain metals reaches 75%, or even 100%. Additionally, the risk of Chinese export restrictions on key raw materials hangs like a Sword of Damocles over European industry[60]. The European Parliamentary Research Service offers a useful overview of EU's view on the matter and notes that export restrictions can generate downstream production stoppages[61].

[55] Junker, B. (2025). *Production disruptions rise: China continues to weaponize rare earths*. *GOLDINVEST*. https://goldinvest.de/en/production-disruptions-rise-china-continues-to-weaponize-rare-earths/
[56] Ericsson, M. (2022). Changing locus of mining. *Mineral Economics, 35*, 1–2. https://doi.org/10.1007/s13563-022-00303-9
[57] Eurostat. (2025). *Material flow accounts statistics - material footprints*. European Commission. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Material_flow_accounts_statistics_-_material_footprints
[58] ibid
[59] Mazarei, A., & Leruth, L. (2022). *Green energy depends on critical minerals. Who controls the supply chains?* Harvard Law School Forum on Corporate Governance. https://corpgov.law.harvard.edu/2022/10/11/green-energy-depends-on-critical-minerals-who-controls-the-supply-chains/
[60] European Parliamentary Research Service underscores the following in a relevant policy paper titled China's rare-earth export restrictions "In response to United States tariffs, and citing national security interests as a reason, China introduced two waves of export controls for rare-earth elements (REEs) in April and October 2025 respectively. The second wave has been suspended until November 2026. The EU has been negatively affected by the controls on REEs, which are indispensable for its digital, green and defence industries." Source: Szczepański, M. (2025). *China's rare-earth export restrictions* (At a glance). European Parliamentary Research Service (EPRS), European Parliament. https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA%282025%29779220
[61] Szczepański, M. (2025). China's rare-earth export restrictions (At a glance, PE 779.220). European Parliamentary Research Service, European Parliament. https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA(2025)779220

As supply chain disruption, triggered by Russia's war in Ukraine, the COVID-19 pandemic, and the escalating US-China rivalry, have heightened global concerns over critical raw materials supply, the EU has come to recognise its own vulnerability[62]. In her **2022 State of the Union** address, European Commission President **Ursula von der Leyen** brought these concerns to the forefront, **stressing the need to diversify supply chains[63]**.

That same year (2022), the United States and several allies launched the **Minerals Security Partnership (MSP)[64]**. The initiative aims to diversify supply chains and reduce dependence on China in the critical-minerals domain, and it has been described as a "**NATO for minerals**". The EU participates in the MSP, though rather than seeking full "**decoupling**" from China, the European strategy is framed around "**de-risking**": reducing reliance on a single supplier state while keeping channels of communication and cooperation open where possible. Finally, the EU's main effort to lessen dependence on external suppliers of critical raw materials is the **Critical Raw Materials Act**.

More specifically, in **March 2023**, Europe's response to critical raw materials shifted from a largely declaratory agenda to a more operational one with the introduction of the **Critical Raw Materials Act**[65]. This legislative initiative sets ambitious targets for **2030**, with four core benchmarks:

a) at least **10%** of the EU's annual consumption of strategic raw materials should be **extracted** within the EU,
b) at least **40%** of the EU's annual consumption of strategic raw materials should be **processed** within the EU,
c) at least **25%** of strategic raw materials should come from **recycling** within the EU, and
d) no single strategic raw material should be sourced **more than 65%** from any third country.

These targets aim to build a resilient supply system capable of absorbing disruptions without severe consequences for the European economy. At the same time, the European Commission made several key assumptions. First, it acknowledged the EU's **high dependence on imports**, often from third countries with **monopoly-like positions**, and the need to reduce the risks associated with that dependence[66]. However, while the EU aims to strengthen domestic supply chains by expanding extraction, processing, and recycling by 2030, the Commission recognised that Europe **cannot be fully self-sufficient** and will continue to rely on imports. For this reason, the EU is pursuing **strategic partnerships** with third-country producers[67].

Additionally, under the new legislation, EU member states are expected to develop national programmes to map and explore their geological resources. Projects designated as "strategic" will benefit from improved access to financing opportunities and faster permitting timelines: **27 months** for extraction projects and **15 months** for processing and recycling[68]. The **Critical Raw Materials Act** entered into force in May 2024 and in 2025, the Commission adopted an initial list of **47 Strategic Projects** inside the EU spanning extraction, processing, and recycling[69].

[62] European Commission, Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs. (2023). *Study on the critical raw materials for the EU 2023: Final report*. Publications Office of the European Union. https://op.europa.eu/en/publication-detail/-/publication/57318397-fdd4-11ed-a05c-01aa75ed71a1
[63] European Commission (2022). Critical Raw Materials Act: Securing the new gas & oil at the heart of our economy [Statement]. European Commission. https://ec.europa.eu/commission/presscorner/detail/mt/STATEMENT_22_5523
[64] U.S. Department of State. *Minerals Security Partnership*. Retrieved January 10, 2026, from https://2021-2025.state.gov/minerals-security-partnership/
[65] European Commission. (2023). *Critical raw materials: Ensuring secure and sustainable supply chains for EU's green and digital future*. European Commission. https://ec.europa.eu/commission/presscorner/detail/en/ip_23_1661.
[66] ibid
[67] ibid
[68] ibid
[69] European Commission. (2025). Commission selects 47 strategic projects to secure and diversify access to raw materials in the EU [Press release]. https://ec.europa.eu/commission/presscorner/detail/en/ip_25_864

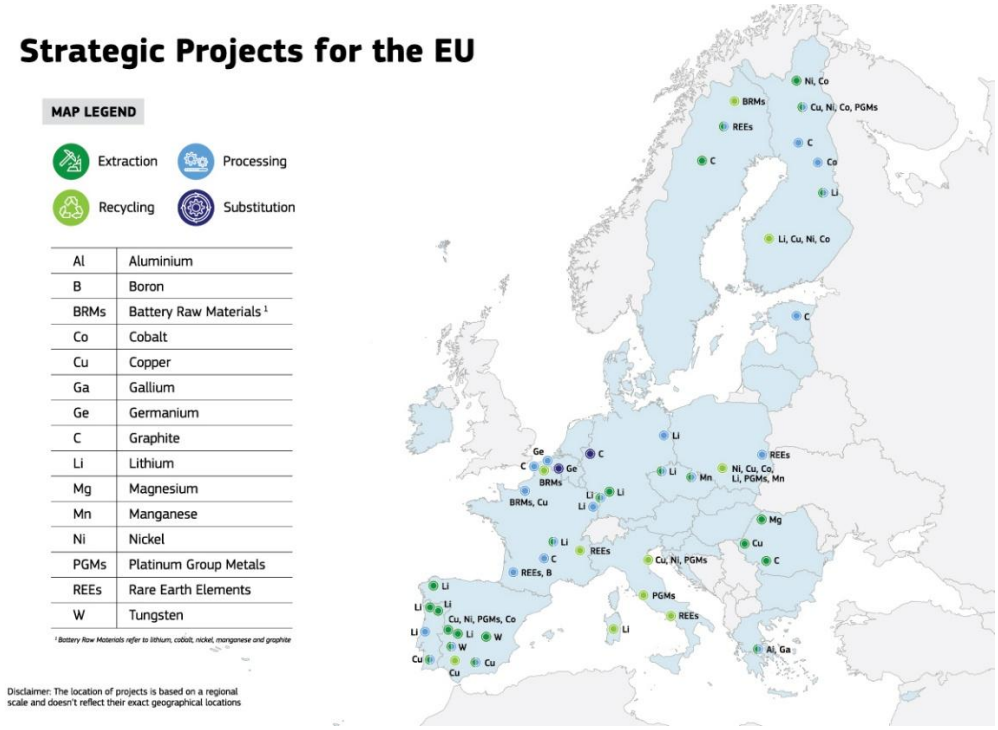


Figure 8: The 47 Strategic Projects *inside the EU*[70]

The EU then extended the same logic externally by adopting, in June 2025, a first list of **13 Strategic Projects located outside the EU** (including a project in Greenland)[71]. As of January 2026, the second call for Strategic Projects had closed, attracting more than 160 applications, including 95 from within the EU and 66 from outside it, indicating continuous momentum in the development of Europe's critical raw materials value chain[72].

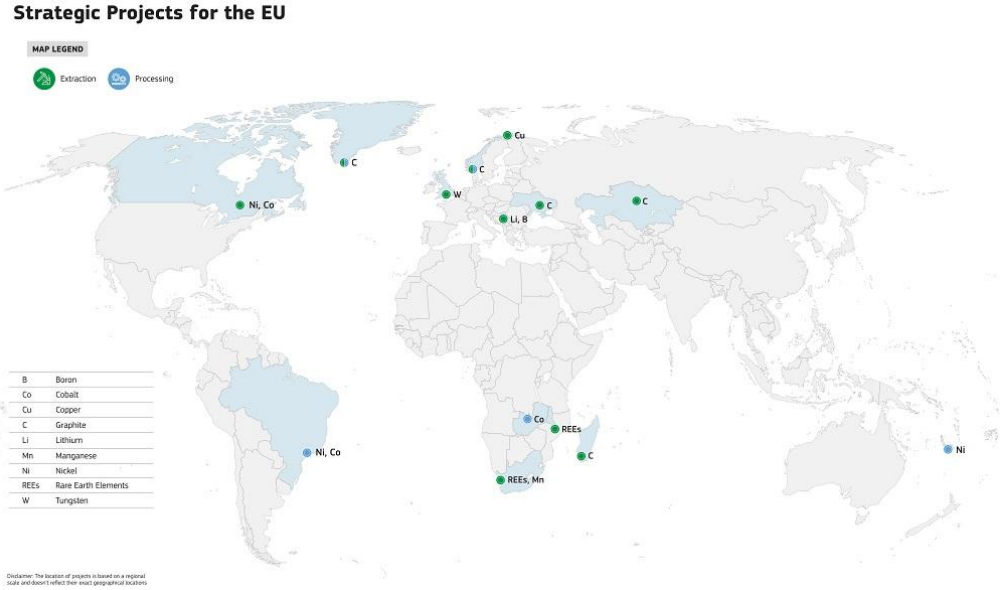


Figure 9: The 13 Strategic Projects located outside the EU[73]

[70] ibid
[71] European Commission. (2025). Commission selects 13 Strategic Projects in third countries to secure access to raw materials and to support local value creation [Press release]. https://ec.europa.eu/commission/presscorner/detail/en/ip_25_1419
[72] European Commission, Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs. (2026). *Strategic projects on critical raw materials gain momentum in second selection round for potential funding and faster permitting*. Internal Market, Industry, Entrepreneurship and SMEs. https://single-market-economy.ec.europa.eu/news/strategic-projects-critical-raw-materials-gain-momentum-second-selection-round-potential-funding-and-2026-01-19_en
[73] ibid

In essence, we can see that the EU is adopting a twofold approach. The first part focuses on strengthening domestic production. Although Europe has not traditionally been a major mining producer, the EU's new strategy introduces incentives to expand the **extraction** and **processing** of critical raw materials within European borders. This includes new mining and metallurgy investments, as well as reinforcing the circular economy through increased recycling[74].

Additionally, apart from the CRMA's Strategic Projects framework, the EU has also supported downstream industrial capacity through **other financial instruments**. For example, in September 2025, Europe's largest rare-earth magnet factory opened in Narva, Estonia, with support from the Just Transition Fund. The significance of the project lies in the fact that it strengthens European manufacturing capacity in rare-earth magnets, one of the segments in which the Union remains highly dependent on Chinese supply[75].

Beyond domestic production, the European strategy also focuses on **diversifying imports through alliances and international partnerships**. The EU is seeking to broaden its supply via agreements with other countries and regions rich in raw materials, including parts of **Africa** and **Latin America**. Europe's external approach relies on partnerships that combine raw materials with infrastructure, energy, and local value creation. For example, in Africa, the Global Gateway frame is meant to differentiate Europe from a purely extractive model by bundling investments into connectivity and industrial capacity. In numbers, **Global Gateway** has seen 138 flagship projects adopted between 2023 and 2025 under the Africa–Europe Global Gateway agenda[76]. The EU also seeks to strengthen its bargaining power by building partnerships with countries with which it maintains strong relations, such as **Canada** and **Australia**. Canada brings processing know-how, particularly in **rare earths**, while Australia has substantial reserves of **lithium** and **rare earth elements**[77].

By late 2025, the European Commission took a further step to speed up investment by enhancing the Critical Raw Materials Act. It put forward the **RESourceEU**[78], an action plan intended to intensify efforts to secure supplies of critical raw materials for strategically exposed value chains, including automotive manufacturing, industrial motors, defence and aerospace industries, and key digital infrastructures such as AI chips and data centres. The initiative is funded by the Critical Raw Materials Act (3 billion euros) and structured around three complementary pillars.

First, it seeks to reduce vulnerability to geopolitical disruption and price volatility by establishing a **European Critical Raw Materials Centre** (planned for the second semester of 2026), supported by the Raw Materials Platform, to aggregate demand, facilitate joint purchasing and offtake arrangements, and develop a more coordinated EU approach to strategic stockpiling. Second, RESourceEU aims to accelerate upstream and midstream projects by de-risking investments and shortening permitting timelines. Third, the plan emphasises on external diversification through deeper cooperation with "like-minded" partners, building on existing strategic partnerships with resource-rich countries.

[74] Payne, J., & Parodi, A. (2025). EU industry chief says raw materials recycling is solution to China dependency. Reuters. https://www.reuters.com/business/retail-consumer/eu-industry-chief-urges-critical-minerals-stockpiling-2025-11-19/
[75] European Commission, Directorate-General for Regional and Urban Policy. (2025). *EU funded Europe's largest rare-earth magnet factory opens in Narva, Estonia*. *Inforegio*. https://ec.europa.eu/regional_policy/whats-new/newsroom/19-09-2025-eu-funded-europe-s-largest-rare-earth-magnet-factory-opens-in-narva-estonia_en
[76] Directorate-General for International Partnerships. (2025). Global Gateway in Africa: European Union and African Union take stock of significant progress. European Commission. https://international-partnerships.ec.europa.eu/news-and-events/news/global-gateway-africa-european-union-and-african-union-take-stock-significant-progress-2025-05-21_en
[77] The Economist Intelligence Unit. (2023). *EU acts to secure access to critical raw materials*. The Economist Intelligence Unit. https://www.eiu.com/n/eu-acts-to-secure-access-to-critical-raw-materials.
[78] European Commission, Directorate-General for Communication. (2025). *New measures to secure raw materials and strengthen the EU's economic security*. European Commission. https://commission.europa.eu/news-and-media/news/new-measures-secure-raw-materials-and-strengthen-eus-economic-security-2025-12-03_en

The Limits of the EU's Critical Raw Materials Strategy

A recent institutional assessment of the EU's efforts in this field by the **European Court of Auditors** provides a revealing picture. In its **Special Report 04/2026**[79], they argue that, while the Critical Raw Materials Act sets a strategic course, its targets lack justification, the underlying data and methodology remain incomplete, and the actual effects of EU funding (which is fragmented across different instruments and directorates) are not properly tracked and documented. The report also finds that efforts to diversify imports have yet to produce tangible results, that strategic partnerships contribute little to securing supply, and that progress in domestic production continues to be constrained by underdeveloped exploration, shrinking processing capacity, financing difficulties, and lengthy permitting procedures. Most importantly, the Special Report warns that many strategic projects are unlikely to secure meaningful supply for the EU by 2030, especially in cases where projects remain in early stages or lack agreements with EU-based customers.

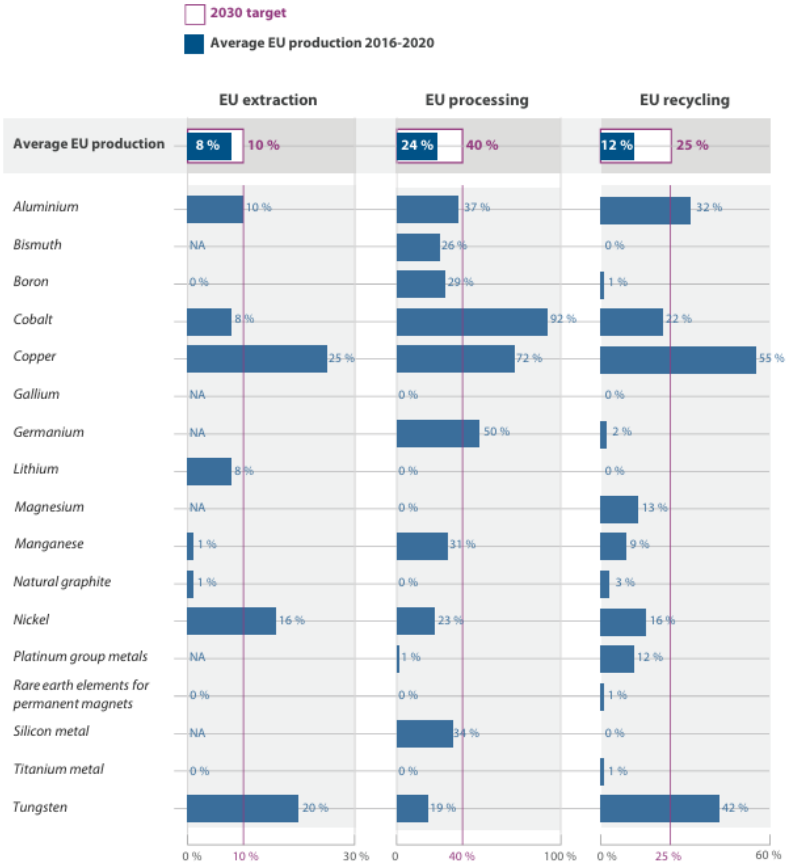
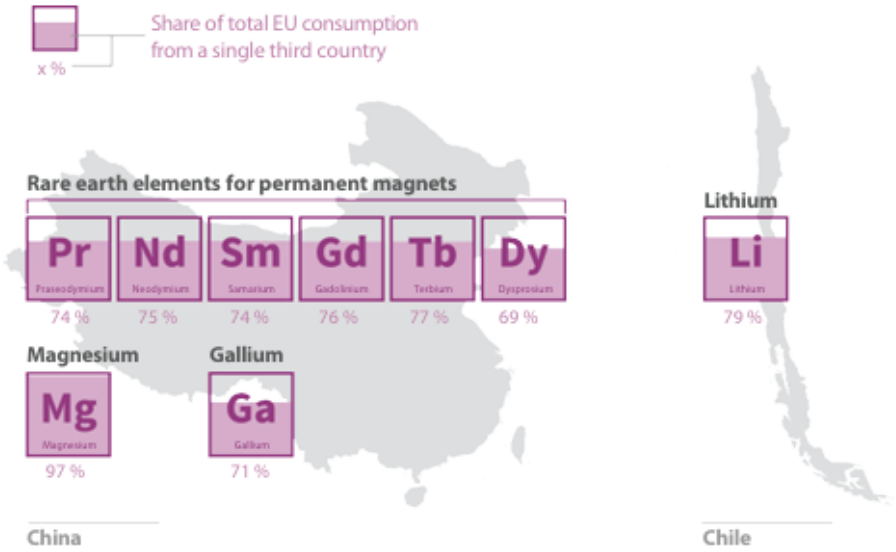


Figure 10 - Average EU production capacity and progress towards 2030 targets[80]

[79] European Court of Auditors. (2026). Critical raw materials for the energy transition: Not a rock-solid policy (Special Report No. 04/2026). Publications Office of the European Union. https://doi.org/10.2865/5008580

[80] The present graph is extracted from the European Court of Auditors (2026) report.



Source: ECA, based on DG GROW data from January 2025.

Figure 11 - EU processed strategic raw materials dependency from single countries above 65%[81]

Europe does not lack strategies, targets, or legislative instruments. The problem is a widening mismatch between the speed at which economic coercion can be applied through price shocks, and targeted export restrictions, and the pace of Europe's reaction, which still runs on multi-year permitting, politically contested projects, and fragmented capital markets.

This interacts directly with what can be described as Europe's environmental paradox. CIRAN[82] research suggests that roughly 85% of Europe's critical-mineral deposits lie inside environmentally protected areas or within close proximity (within 5 km), creating barriers to domestic extraction. The constraint becomes sharper when combined with the Nature Restoration Regulation's requirement that restoration measures cover at least 20% of EU land and sea by 2030[83]. Additionally, the intended supply-chain diversification faces serious constraints. Sourcing minerals from outside the EU brings its own complications, especially as EU rules on **Environmental, Social, and Governance (ESG)** standards become stricter. At present, there is no fully standardised way to verify whether raw materials imported from third countries have been extracted in an environmentally sustainable and socially responsible manner. Moreover, the success of Europe's strategy also depends on societal acceptance. The **Serbian Jadar lithium** project stands as a clear example, where continued public resistance can erode legitimacy, especially when projects are framed as serving external needs while local communities internalise environmental risk[84].

Lastly, **Europe's capital markets are fragmented**. That makes it hard to move large amounts of money towards EU priorities such as raw materials projects, reindustrialisation, and competitiveness. The problem is especially acute in mining, because the pool of investors able to finance minerals is small[85]. The aforementioned restriction coupled with **delayed timelines** (e.g. the battery due diligence obligations under the Batteries Regulation were postponed to 18 August 2027) illustrates **Europe's shortfalls**. Regardless of its insistence on high standards, the administrative capacity, bureaucracy, corporate readiness, and current infrastructure may act as clear obstacles to its wider strategic aspirations.

[81] The present graph is extracted from the European Court of Auditors (2026) report.
[82] CIRAN (CrItical RAw materials extraction in enviroNmentally protected areas) is an EU-funded Horizon Europe project focused on developing policies that balance the need for critical raw materials with the protection of sensitive ecosystems.
[83] Correia, V., & Falck, W. E. (2025). Europe's critical raw materials: Balancing strategic needs with environmental protection. *Intereconomics: Review of European Economic Policy, 60*(5), 297–301. https://doi.org/10.2478/ie-2025-0057
[84] Reuters. (2025). Serbian farmers vow to oppose Rio Tinto lithium project even after EU labels it strategic. *Reuters*. https://www.reuters.com/business/serbian-farmers-vow-oppose-rio-tinto-lithium-project-even-after-eu-labels-it-2025-06-05/
[85] Correia, V., & Falck, W. E. (2025). Europe's critical raw materials: Balancing strategic needs with environmental protection. *Intereconomics: Review of European Economic Policy, 60*(5), 297–301. https://doi.org/10.2478/ie-2025-0057

This is where the comparison with China becomes unavoidable. It is structurally difficult for a rule-of-law, pluralist polity to compete with an authoritarian political economy that can mobilise capital, coordinate state and private companies, compress timelines, neglect environmental standards, and accept costs that European societies are far less willing to bear. Undeniably, the issue at stake is not whether the EU should abandon its democratic foundations and adopt an authoritarian playbook to advance its interests. It is whether it can build institutions capable of reconciling environmental, social, and legal standards with strategic execution. The difficulties surrounding the EU-Mercosur agreement offer a revealing example of how strategic initiatives can become hostage to national preferences and domestic political competition. In an era in which new trade agreements and strategic partnerships are being forged constantly, while long-standing alliances and diplomatic practices are being challenged, the real question is whether the EU can act with sufficient agility and flexibility to secure deals with new suppliers that remain outside the Chinese and even American radar.

The emerging role of Greece in the raw materials sector

Greece could emerge as a meaningful contributor to the EU's effort to reduce strategic dependence on critical raw materials, not only because of its mineral endowment, but because it may help alleviate specific vulnerabilities within European supply chains.

The country possesses deposits of nickel and magnesite and is already a significant producer of bauxite, a key input for aluminium production. In addition, research projects are under way to assess the feasibility of extracting **lithium** and **rare earth elements**, which are vital for electric-vehicle batteries and renewable energy systems[86]. At the same time, the private company Rockfire Resources identified germanium[87] at the Molai deposit in Laconia[88], one of the rarest critical metals globally, and plans to move forward with mining operations[89]. In 2024, it also announced that the Molai deposit contains 1,090,000 tonnes of zinc, 260,000 tonnes of lead, and 19.1 million ounces of silver[90].

It is also worth noting that the Greek government is expected to launch relevant tenders for the exploitation of **graphite deposits** in **Thermes**[91] and **manganese oxide** in **Drama**[92]. In this context, on **12 December 2024**, Greece amended **Law 4864/2021** in order to promote and facilitate a new category of investments**, "flagship investments of exceptional importance"**. This category includes, among other areas, projects related to **critical raw materials**[93]. Designation as a flagship investment provides a range of advantages, including fast-track approvals within **45 days**, location-based incentives, cost-support measures, and tax benefits. These tax benefits include stabilising corporate income tax for up to **12 years**, exemptions from income tax payments, and accelerated depreciation[94].

A key player in the domestic market is Metlen Energy and Metals. One of Metlen's projects, developed as a joint venture with European Bauxites, is included in the Commission's list of **47 Strategic Projects** and covers **bauxite mining** as well as the production of **alumina/aluminium** and the recovery of **gallium** in Boeotia. This strategic

[86] CET Editor. (2025). *EU looks to Greece for critical raw materials*. Central European Times. https://centraleuropeantimes.com/2025/02/eu-looks-to-greece-for-critical-raw-materials.
[87] Germanium is used in the manufacture of mobile phones, electronics, solar cells, camera lenses, satellites, computer monitors, as well as steering and parking sensors for vehicles. It is also used in numerous military applications, including weapons sights and infrared night vision.
[88] Rockfire Resources plc. (2024, September 4). *500% increase in Resource at Molaoi* [RNS announcement]. *Financial Times Markets Data*. https://markets.ft.com/data/announce/full?dockey=1323-16648690-3LTU1E4IS1T6MR3ELE5S8S0RFC
[89] CET Editor. (2025). *EU looks to Greece for critical raw materials*. Central European Times. https://centraleuropeantimes.com/2025/02/eu-looks-to-greece-for-critical-raw-materials.
[90] Avaris (2024). *Μολάοι: Συνεχίζονται οι ερευνητικές γεωτρήσεις το 2025*. EnergyGame. https://www.energygame.gr/protes-yles/429659/molaoi-synechizontai-oi-erevnitikes-geotriseis-to-2025.
[91] In June 2025, EAGME submitted to the Ministry of Environment and Energy (YPEN) an updated dossier on the investment's viability, with the aim of initiating the preparatory procedures.
[92] CET Editor. (2025). *EU looks to Greece for critical raw materials*. Central European Times. https://centraleuropeantimes.com/2025/02/eu-looks-to-greece-for-critical-raw-materials.
[93] United Nations Conference on Trade and Development (UNCTAD). (2024). *Greece promotes investments in critical raw materials, the circular economy, and shipbuilding*. UNCTAD. https://investmentpolicy.unctad.org/investment-policy-monitor/measures/4923.
[94] ibid

significance became more concrete on 15 January 2026, when the European Investment Bank approved €90 million in financing to METLEN for investments in Central Greece. According to the EIB, the project includes the modernisation of bauxite mining operations in the Parnassus–Giona area and the development of a new gallium production facility at the Aluminium of Greece complex in Agios Nikolaos, Viotia[95].

This is important as it links Greece not only to extraction, but also to processing, the very stage at which Europe remains exposed. During a visit to the company's aluminium facilities, Executive Vice-President of the European Commission for Prosperity and Industrial Strategy **Stéphane Séjourné** stated that the project "could account for **10% of our aluminium needs** and might be able to **cover EU demand for gallium**"[96]. Moreover, Evangelos Mytilineos, Executive Chairman of Metlen Energy & Metals is the President of Eurometaux, Europe's Association for the Non-Ferrous Metals industry[97]. This further highlights Greece's emerging position at the European industrial level.

Finally, it is worth emphasising that beyond extraction itself, Greece's strategic relevance may derive from its proximity to major global trade routes, combined with developed ports and supply networks. This creates the conditions for the country to act not only as a mining location, but also as a potential **processing and export hub**. Provided that the necessary investment, permitting capacity, and industrial coordination are secured, Greece could contribute to the EU's effort to reduce strategic vulnerabilities in critical raw material supply chains.

Strategic Priorities for Reducing Europe's Critical Raw Materials Vulnerability

Since Europe's critical raw materials exposure is located across several parts of the value chain, there is not a single policy proposal, which can act as a holistic remedy. The main target shall be the reduction of **coercible concentration** at stages where disruptions or delays would quickly affect industrial capacity. This implies a multifaceted approach with a plethora of measures, which shall reinforce each other. At the same time, we should make clear that Europe can neither entirely exclude China as a supplier, nor sustain the current model.

First, the Union must widen its strategic horizon**. Deeper engagement with MERCOSUR, Africa, Asia[98] and the Western Balkans[99] will be essential for accessing more diversified sources and routes of supply of raw materials**. This orientation stands as a structural necessity, as the United States itself is prioritising reshoring, supply-chain resilience, and industrial autonomy through a more protectionist trade posture[100].

Second, Europe should diversify its supply chains following an **investment-led approach**. Transport corridors, port and rail logistics are an appropriate way of advancing its interests and securing the desired supply chains. For example, the Commission's Joint Research Centre has assessed **Global Gateway** support to transport corridors as central for value chains[101]. Nevertheless, all the above should materialise in a mutually beneficial way for both EU and third countries, as securing access to raw materials could reproduce older, colonial-style patterns rendering some third countries as "Green colonies".

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[95] European Investment Bank. (2026, January 15). *EIB backs Europe's first gallium production investment with €90 million financing to METLEN in Greece*. https://www.eib.org/en/press/all/2026-011-eib-backs-europe-s-first-gallium-production-investment-with-eur90-million-financing-to-metlen-in-greece

[96] Voudouri, A. (2025). *Greece could provide 'totality' of EU gallium demand*. eKathimerini.com. https://www.ekathimerini.com/economy/1265760/greece-could-provide-totality-of-eu-gallium-demand/

[97] METLEN Energy & Metals. (2024). *METLEN's Chairman & CEO Evangelos Mytilineos re-elected as President of Eurometaux* [Press release]. https://www.metlengroup.com/news/press-releases/metlen-s-chairman-ceo-evangelos-mytilineos-re-elected-as-president-of-eurometaux/

[98] Åberg, A., & Townend, R. (Hosts). (2025, April 8). *The geopolitics of critical raw materials* [Audio podcast episode]. In *The Climate Briefing*. Chatham House. https://www.chathamhouse.org/2025/04/climate-briefing-geopolitics-critical-raw-materials

[99] Krstinovska, A., & Wolf, A. (2025). Raw materials for a resilient Europe: The EU's strategic partnership with the Western Balkans. ELIAMEP. https://www.eliamep.gr/en/raw-materials-for-a-resilient-europe-the-eus-strategic-partnership-with-the-western-balkans/

[100] Kalantzakos, S. (2026). Caught in the middle: Industrial power, strategic minerals, and Europe's hard geopolitical choices. IEP@BU, Institute for European Policymaking at Bocconi University.

[101] European Commission, Joint Research Centre. (2025). *Global Gateway support to transport corridors in Africa: Consolidated report* (EUR 40431; JRC142789). Publications Office of the European Union.

Thirdly, the most consequential priority is to expand capacity in **mining**, **processing, conversion, and intermediates**. This also derives directly from the EU's self-diagnosis under the Critical Raw Materials Act, which sets targets not only for extraction but also for processing and recycling, and seeks to reduce reliance on any single third country beyond a defined threshold. This priority is further underscored by the European Court of Auditors' Special Report, which identifies lack of technology, shrinking processing capacity, financing difficulties, and lengthy permitting as central obstacles[102].

A fourth priority is to increase Europe's **contracting power**, which addresses a different structural weakness, fragmented demand and uncoordinated procurement. When firms procure separately and under uncertainty, it is harder to finance new projects and easier for dominant actors to shape the market. The **EU Energy and Raw Materials Platform**[103], launched in July 2025, and the upcoming **European Critical Raw Materials Centre** (planned for 2026[104]) aims to collect demand and match it to supply offers, with the possibility of joint purchasing and ventures.

The fifth proposal, which complements the previous two, is to strengthen the emerging **European Critical Raw Materials Centre** into a cross-instrument coordinating body. Its purpose should not be to add another layer of governance, but to reduce the fragmentation that currently weakens the EU's CRM strategy by aligning funding across instruments and directorates, tracking project-level outcomes, and linking strategic partnerships to concrete roadmaps and financing structures. Such a body would mitigate one of the EU's clearest weaknesses identified by the European Court of Auditors' Special Report[105].

The last proposal treats **circularity** as a security-relevant supply policy. This includes scaling collection and recycling capacity. The **RESourceEU** Action Plan includes measures to strengthen recycling and proposes restrictions on exports of scrap and waste of permanent magnets, alongside targeted measures for aluminium scrap. The European Court of Auditors' Special Report similarly finds that the EU does not yet make full use of substitution, recycling, and sustainable resource management[106]

In conclusion, **Europe** does not lack strategy or legal instruments. The key risks for Europe are its heavy reliance on a small number of actors in processing and key intermediates, exposure to export restrictions, limited room to expand domestic mining, absence of a coordinating body that could support partnerships, tied to concrete supply outcomes, and supervise funding, and fragmented capital markets that struggle to finance projects. If these risks are not addressed, Europe may face higher costs, supply disruptions, and strategic weakness across sectors like automotive, energy, defence, and advanced manufacturing.

For **Greece**, the task is also challenging. The risks are predominantly political and social. Public support can erode quickly if projects are perceived as externally driven, environmentally costly, or opaque. There is also a growing risk that Greece becomes a "pass-through" corridor, moving materials through its ports, without capturing added value. This is why it is important to anchor processing, skills, and industrial upgrading domestically. At the same time, Greece can position itself as a credible contributor to EU supply security by acting as a processing and industrial node linked to European value chains that supports diversified supply routes.

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[102] European Court of Auditors. (2026). Critical raw materials for the energy transition: Not a rock-solid policy (Special Report No. 04/2026). Publications Office of the European Union. https://doi.org/10.2865/5008580

[103] European Commission. (2025). EU Energy and Raw Materials Platform. https://energy.ec.europa.eu/topics/energy-security/eu-energy-and-raw-materials-platform_en

[104] European Parliament. (2026). *Critical Raw Materials Centre*. *Legislative Train Schedule*. https://www.europarl.europa.eu/legislative-train/theme-a-new-plan-for-europe-s-sustainable-prosperity-and-competitiveness/file-critical-raw-materials-centre

[105] European Court of Auditors. (2026). Critical raw materials for the energy transition: Not a rock-solid policy (Special Report No. 04/2026). Publications Office of the European Union. https://doi.org/10.2865/5008580

[106] ibid

ANNEX

The 34 key critical raw materials according to the European Commission (2023) [107]. The 17 strategic raw materials are in *italics,* as they constitute a subset of the 34 key critical raw materials.

Raw materials	Main global producers	Main EU sourcing countries[108]	Import reliance[109]	Selected Uses
Aluminium/ bauxite	Australia 28% China 21% Guinea 18%	Guinea 62% Brazil 12% Greece 10%	89%	Lightweight structures High-tech engineering
Antimony	China 56% Tajikistan 20% Russia 12%	Türkiye 63% Bolivia 26% China 6%	100%	Flame retardants Defence applications Lead-acid batteries
Arsenic	China 44% Peru 40% Morocco 11%	Belgium 60% China 39%	39%	Semiconductors Alloys
Baryte	China 32% India 25% Morocco 9%	China 44% Morocco 28% Bulgaria 11% Germany 7% Slovakia 2%	74%	Medical applications Radiation protection Chemical applications
Beryllium	United States 67% China 26% Mozambique 4%	n/a	n/a	Electronic and Communications Equipment automotive, aero-space and defence components
Bismuth	China 70% Vietnam 18% Japan 5%	China 50% Belgium 26% Thailand 9% Laos 5%	71%	Pharmaceuticals Medical applications Low-melting point alloys Solid rocket propellant
Boron	Türkiye 48% United States 25% Chile 11%	Türkiye 99%	100%	High performance glass Fertilisers Permanent magnets
Cobalt	Congo, D.R. 63% Russia 7% Canada 4%	n/a	81%	Batteries Super alloys Catalysts Magnets
Coking coal	China 53% Australia 18% Russia 9% United States 6%	Poland 26% Australia 24% United States 20% Russia 8% Canada 5% Czechia 5% Germany 2%	66%	Coke for steel Carbon fibres Battery electrodes
Copper	Chile 28% Peru 12% China 8%	Poland 19% Chile 14% Peru 10%	48%	Electrical infrastructure

[107] European Commission, Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs. (2023). *Study on the critical raw materials for the EU 2023: Final report*. Publications Office of the European Union. https://op.europa.eu/en/publication-detail/-/publication/57318397-fdd4-11ed-a05c-01aa75ed71a1

[108] Based on domestic production and imports (excluding exports)

[109] Import Reliance = (Import – Export) / (Domestic production + Import – Export)

		Spain 8% Bulgaria 5% Sweden 4% Finland 2% Portugal 2%		
Feldspar	Türkiye 32% India 20% China 8% Italy 7%	Türkiye 51% Italy 22% Spain 7% France 5% Czechia 4% Germany 2% Portugal 1% Poland 1%	54%	Glass including fibreglass Ceramics
Fluorspar	China 56% Mexico 21% Mongolia 7%	Mexico 33% Spain 23% South Africa 11% Germany 8%	60%	Steel and iron making Refrigeration and Air-conditioning Aluminium making and other metallurgy
Gallium	China 94% Ukraine 2% Russia 2%	China 71% United States 10% United Kingdom 9%	98%	Semiconductors Photovoltaic cells
Germanium	China 83% Russia 5% Belgium 4% Germany 3% Japan 2% United States 2%	China 45% Belgium 32% Germany 19%	42%	Optical fibres and Infrared Optics Satellite solar cells Polymerisation catalysts
Hafnium	France 76% Ukraine 14% China 5% Russia 3%	France 49% United States 44% Russia 3%	0% (EU is a net exporter of Hafnium)	Super alloys Nuclear control rods Refractory ceramics
Helium	United States 56% Qatar 30% Algeria 8%	Qatar 34% Algeria 29% United States 21% Poland 5%	94%	Controlled atmospheres Semiconductors MRI
Lithium	China 56% Chile 32% Argentina 11%	Chile 79% Switzerland 7% Argentina 6% United States 5%	100%	Batteries Glass and ceramics Steel and aluminium metallurgy
Magnesium	China 91% United States 3%	China 97% Israel 1%	100%	Lightweight alloys for automotive, electronics, packaging or construction Desulphurisation agent in steelmaking
Manganese	South Africa 29% Australia 16% Gabon 14%	South Africa 41% Gabon 39%	96%	Steel-making Batteries

	China 9%	Brazil 8% Ukraine 3%		
Natural Graphite	China 67% Brazil 8% Mozambique 5% India 5% Korea, North 5%	China 40% Brazil 13% Mozambique 12% Norway 8% Ukraine 7%	99%	Batteries • Refractories for steelmaking
Nickel	China 33% Indonesia 12% Japan 9% Russia 7% Canada 6% Australia 5%	Russia 29% Finland 17% Norway 10% Canada 6% Australia 6%	75%	Batteries Steel making Automotive
Niobium	Brazil 92% Canada 7%	n/a	100%	High-strength steel and super alloys for transportation and infrastructure High-tech applications (capacitors, superconducting magnets, etc.)
Phosphate rock	China 44% Morocco 14% United States 10% Russia 7%	Morocco 27% Russia 24% Finland 17% Algeria 10%	82%	Mineral fertilizer Phosphorous compounds
Scandium	China 67% Russia 17%	n/a	100%	Solid Oxide Fuel Cells Lightweight alloys
Silicon metal	China 76% Brazil 7% Norway 6% France 4%	Norway 33% France 27% Brazil 9%	60%	Semiconductors Photovoltaics Electronic components Silicones
Strontium	Iran 37% Spain 34% China 16%	Spain 99%	0%	Ceramic magnets Aluminium alloys Medical applications Pyrotechnics
Tantalum	Congo, D.R. 35% Rwanda 17% Brazil 16% Nigeria 11%	99%	n/a	Capacitors for electronic devices Super alloys
Titanium metal	China 25% South Africa 13% Australia 12% Mozambique 10% Canada 8% Ukraine 6%	Kazakhstan 36% Russia 34% China 9% Switzerland 5%	100%	Lightweight high-strength alloys for e.g. aeronautics, space and defence Medical applications
Tungsten	China 86% United States 4% Russia 3% Vietnam 3% Austria 2%	China 31% Austria 19% Vietnam 14% Russia 9%	80%	Alloys e.g. for aeronautics, space, defence, electrical technology Mill, cutting and mining tools
Vanadium	China 62% Russia 20%	n/a	n/a	High-strength-low-alloys for e.g. aeronautics, space,

	South Africa 11% Brazil 8%			nuclear reactors Chemical catalysts
Platinum Group Metals	South Africa (iridium, platinum, rhodium, Ruthenium) 94% Russia (palladium) 40%	n/a	100%	South Africa 94% - iridium, platinum, rhodium, ruthenium Russia - palladium 40%
Heavy Rare Earth Elements	China 100%	n/a	100%	Permanent Magnets for electric motors and electricity generators Lighting Phosphors Catalysts Batteries Glass and ceramics
Light Rare Earth Elements	China 85% Malaysia 11%	n/a	100%	